

Neuropharmacology Part I

Lecture for Neuro and behavioral
science module

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Learning Objectives

- Understand how drugs act on central nervous system
- Understand roles of central depressants
- Dangers of central depressants

Reference materials

- Basic and Clinical Pharmacology, Katzung
- Goodman and Gillman, the Pharmacological basis of therapeutics
- Pharmacology, Rang and Dale
- Integrated Pharmacology, Page, Curtis, Sutter, Walker and Hoffman

Anatomy and physiology of neurons

- Human brain- very complex
- Composed of billion neurons with functional interdependence
- Mixtures of excitatory and inhibitory neurons
- Excitation is brought by
 - Activation of excitatory neurons
 - Blockade of inhibitory neurons
- Inhibition is accomplished by
 - Pre-synaptic inhibition
 - Postsynaptic inhibition
- Excitation/inhibition exhibited by specific transmitters

Neurotransmitters in the brain

- Acetylcholine: excitatory (M_1 and N receptors)
- Monoamines
 - NA: excitatory (α_1), inhibitory α_2)
 - Dopamine: inhibitory (D_1 , D_2)
 - 5HT: inhibitory (5HT-1A), excitatory (5HT_{2A}, 5HT₃, 5HT₄)
 - Histamine: excitatory (H_1 , H_2), inhibitory (H_3)
- Amino acids
 - Neutral
 - Glycine: inhibitory at spinal level
 - GABA: inhibitory at brain level
 - Acidic
 - Glutamate/aspartate: excitatory
- Peptides
 - Enkephalins, endorphins: inhibitory
 - Substance P: excitatory

Blood brain barrier

- **Structure**
 - Capillaries closely joined
 - Absence of pores
- **Function**
 - Does not allow polar molecules to pass through
 - Protect the brain for peripheral endogenous substances

Sites of CNS drugs action

- AP
- Synthesis
- Storage
- Release
- Reuptake
- Degradation
- Receptors/ion channels

Ranges of effects

Coma ← anesthesia ← hypnosis ← sedation
→ excitation → hyper-excitation → convulsion

Classes of CNS drugs

- General (non-selective) central depressants
 - General anesthetics
 - Sedative-hypnotics
 - Alcohols
- General (non-selective) central stimulants
 - Blockade of inhibition
 - Strychnine
 - Picrotoxin
 - Direct neuronal excitation
 - Metrazole
 - Xanthines
- Selective CNS functions modifiers
 - Psychoactive drugs
 - Opioid analgesics
 - Anti-epileptics
 - Antiparkinsonians

General anesthetics

- Sequence of anesthesia
 - Pain neurons
 - Inhibitory neurons
 - Excitatory neurons
 - Motor neurons

Clinical stages of anesthetics

- Stage I: stage of analgesia/induction
 - Due to inhibition of pain neurons
 - Analgesia
 - Intact consciousness
- Stage II: stage of delirium/excitement
 - Due to inhibition of inhibitory neurons
 - Loss of consciousness
 - Exaggerated reflexes
 - Exaggerated muscle tone
 - Exaggerated sympathetic activity
 - Urine/feces incontinence
 - Irregular breathing

Clinical stages of anesthesia

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 - Irregular breathing

Clinical stages (cont.)

- Stage III: stage of surgical anesthesia
 - Due to inhibition of excitatory neurons

	Plane 1	2	3	4
• Eye ball pos.	Moving	fixed	fixed	fixed
• Pupil size	miosis	1/2myd	3/4myd	full myd
• Reflexes suppressed	+	++	+++	++++
• Skeletal muscle tone	↓	↓ ↓	↓ ↓ ↓	↓ ↓ ↓ ↓
• Respiration	regular/full	regular/ less autonomic	regular with pause	irregular/ /shallow

- Stage IV: Stage of medullary paralysis
 - Due to inhibition of motor neurons
 - Paralysis of respiration

Classes of general anesthetics

- Inhalational
 - Gaseous
 - Nitrous oxide
 - Volatile liquids
 - Diethyl ether
 - Halothane
 - Enflurane
 - Isoflurane
 - Methoxyflurane
- Parentral
 - Basal anesthetics: thiopental, propofol
 - Neurolept analgesics: droperidol + fentanyl
 - Dissociative anesthetics: ketamine

Inhalational anesthetics

- Mechanisms of action
 - Water theory
 - Anesthetic + water → clathrate
 - Lipid theory
 - Anesthetic + hydrophobic tails → membrane fluidity
 - Protein theory
 - Anesthetics + receptors → modified receptors
 - NMDA receptor
 - GABA receptor
 - Glycine receptor

kinetics

$$P_t = p_{p_1} + p_{p_2} + p_{p_3} + \dots + p_{p_n}$$

Affinity: function of solubility

Free gas: exerts pressure

Conc. of gas = $p_p \times \text{affinity}$

Kinetics cont.

Inhalation of anesthetic



Alveolar tension



Arterial tension



brain/tissue tension



venous tension

Factors affecting kinetics

- Pulmonary ventilation
- Partial pressure of anesthetic
- Blood: gas partition coefficient
- Cardiac output (pulmonary blood flow)
- Lipid: gas partition coefficient
- Rate of blood flow

Techniques of administration

- Open drop method
 - Poured over a mask with gauze to be inhaled
 - Lot of anesthetics escapes
 - Used for cheaper anesthetics
- Anesthetic machine
 - Gas cylinder
 - Flow meter
 - Unidirectional valves
 - Delivered through tightly fitted facemask or endothelial tube → precise conc.

Techniques of administration (cont.)

– Open system

- Exhaled gas allowed to escape
- Fresh gas inhaled each time
- No re-breathing → precisely measured

– Closed system

- Re-breathing
- Difficulty in precise measurement
- Useful for expensive anesthetics

– Semi-closed system

- Partially re-breathing with intermediate flow rate

Potency of inhalational anesthetics

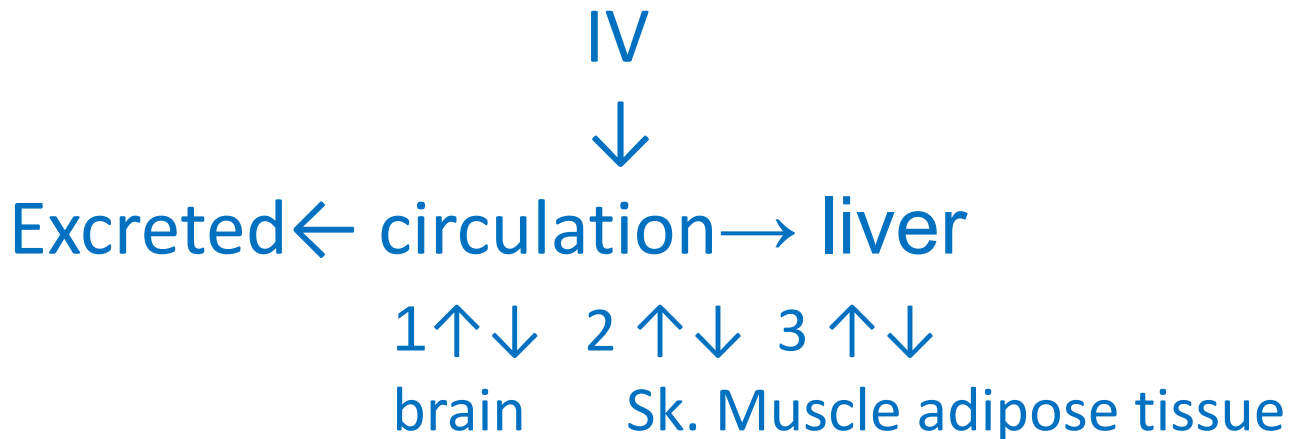
- Depends on lipid: gas partition coefficient
- Expressed as MAC
- MAC=minimum alveolar conc. of anesthetics that produces anesthesia in 50% of the patients

Comparison of Inhalational anesthetics

Property	nitrous oxide	ether	halothane	enflurane	isoflurane
Flammability	no	yes	no	no	no
Irritation	no	yes	no	no	no
PO vomiting	no	yes	no	no	no
B:G part. Coef.	0.47	12	2.5	1.9	1.4
Onset	v. rapid	slow	reasonable	reasonable	reasonabl
Recovery	V. rapid	slow	reasonable	reasonable	reasonabl
O:G part. Coef.	1.4	65	220	98	91
MAC	>100%	1.9%	0.75%	1.7%	1.15%
Potency	V. poor	potent	V. potent	potent	potent
Purpose	induction	maintenance same		same	same
Analgesia	reasonable	strong	poor	poor	poor
Sk. Muscle relaxation	poor	strong	poor	poor	poor
Status	used	not used	used	used	used

basal anesthetics

- Kinetics
 - Given IV



- Eliminated through redistribution
- Eventually cleared by metabolism
- Used for induction and short procedures (burn dressing, set fractures)

Neurolept analgesics

- Used when wakeful conditions required for cooperation

Dissociative anesthetics

- Act by blocking glutamate effect at NMDA receptor
- Bring about dissociation
- Produce post operative hallucinations
- Intact consciousness
- Used in children

Properties of Ideal anesthetics

- have
 - Sedative effects
 - Analgesic effect
 - Muscle relaxant effect
 - Vagolytic effect
 - Non irritant property
 - Rapid induction and reasonable recovery
 - No gastric regurgitation in prolonged surgery
- No anesthetic meets all these requirements

Therefore there is a need to balance anesthesia

Pre-medication

- Sedatives
- Analgesics
- Muscle relaxants
- Atropine like drugs
- Antiemetics
- Basal anesthetics
- H₂ blockers

Hypnotics

- Sedatives: bring about drowsiness/anxiolytic effect
- Hypnotics: bring about sleep/used in insomnia

Causes of insomnia

- Specific like pain
 - No hypnotic is prescribed
- Nonspecific
 - Hypnotics are prescribed

Factors affecting selection of hypnotics

- Nature of sleep disorder
 - Difficulty in falling asleep
 - Rapid acting hypnotics
 - Difficulty in maintaining asleep
 - Long acting hypnotics
 - Difficulty in both
 - Rapid and long acting hypnotics

Factors for determining dose of hypnotics

- Characteristics of patient's daily activity
 - Students vs housewives
- Past experience with medication
 - tolerance
- General patient attitude to drugs
 - Positive
 - Negative

Classes of hypnotics

- Oldest hypnotics
 - Paraldehyde, urethane: no more used
- Barbiturates
 - Ultrashort acting
 - Thiopental
 - Methohexital
 - Thiamylal
 - Short to intermediate acting
 - Pentobarbital
 - Secobarbital
 - Hexobarbital
 - Long acting
 - Phenobarbital

Classes of hypnotics (cont.)

- Benzodiazepines

- Short acting

- Oxazepam
 - Triazolam
 - Lorazepam
 - Alprazolam

- Long acting

- Diazepam
 - Temazepam
 - Flurazepam
 - Chlorazepam

Classes of hypnotics (cont.)

- Newer hypnotics
 - Zolpidem
 - Zaleplon
 - Eszopiclone

Comparison between barbiturates and benzodiazepines

Barbiturates

Benzodiazepines

Mech

↑GABA effect

activate B-receptors

Effects

- sedation
- hypnosis
- anesthesia
- REM phase
- Hangover
- Nightmares
- Rp
- BP
- Addiction

low dose
intermediate dose
high dose
↓ ↓ ↓
more
more
↓ ↓ ↓
↓ ↓ ↓
high

low dose
intermediate dose
high dose
↓
less
less
↓
↓
low

Indications

- sedatives
- hypnotics
- basal anesthetic
- antiepileptic

short acting
short acting
ultra-short acting
long acting

all
all
diazepam
diazepam

Alcohols

- Have been used for social and medical purposes
- Obtained through
 - Brewery
 - Winery
 - distillation
- Arabs were the first to introduce distillation
- Whisky means “Water of Life”

Ethanol

- Kinetics
 - Absorbed both from alveoli and GIT
 - Rapidly distributed
 - Biotransformed through
 - Alcohol dehydrogenase
 - MEOS
 - Excreted in urine and alveoli

Ethanol Cont.

- **Effects**

- ↑acetylcholine release
- ↑ saliva → ↑appetite followed with suppression
- ↑ acid secretion → ↑digestion
- Vasodilatation → ↓BT
- ↓ADH release → micturation
- Impairs mental capacity
- Slurred speech/fluent language
- Avoids self restraint
- Addiction liability

Ethanol

- Indication

- locally as antiseptic/disinfectant
- In methanol poisoning

- Toxicity

- Acute

- S/S

Alcohol smell

Drunkenness

- Treatment

Withdrawal

- Chronic

- S/S

hepatitis

appetite suppression

- Treatment

alcohol antabuse
(disulfiram)

Selective CNS modifiers

Antiepileptic drugs

- Epilepsy: a syndrome characterized by
 - Sudden transient alterations in basic functions leading to
 - Motor
 - Sensory
 - Autonomic
 - Psychic symptoms
 - With or without convulsion
 - Accompanied by unconsciousness or consciousness

Types of epilepsy

- Generalized seizures
 - Grand mal seizure
 - Peti-mal seizure
- Partial seizures
 - Simple
 - Complex

Etiology of epileptic seizures

- Abnormal discharge due to
 - Ca channel activation
- Abnormal spread due to
 - Na channel activation
 - GABA blockade

Objectives of treatment

- Prevention and Control of seizure through
 - Blocking Ca channel
 - Blocking Na channel
 - Potentiating GABA

Mechanisms for preventing and controlling epileptic seizures

- ↓ abnormal discharge through blockade of Ca channel
 - Succinimides: ethosuximide
 - Oxazolidinediones: trimethadone
 - Valproic acid
- ↓ spread of discharge through
 - ↓ Na channels
 - Hydantoins: Phenytoin
 - Valproic acid
 - Primidone , Carbamazepine

Classes of antiepileptic drugs based on Mechanisms of actions....

– ↑GABA

- Barbiturates: phenobarbital
- Benzodiazepines: diazepam, clonazepam
- Valproic acid
- Gabapentin
- Vigabatrine
- Lamotrigine

Classes of antiepileptics based on generations

- First generation
 - Barbiturates and related drugs: phenobarbital, primidone
 - Benzodiazepines: diazepam, clonazepam
 - Carbamazepine
 - Valproic acid
 - Succinimides: ethosuximide
- Second generation
 - Vigabatrin
 - Lamotrigine, Gabapentin

Principles of antiepileptic therapy

- Referral to neurologist as soon as possible
- Diagnosis prior to initiating treatment
- Initiate treatment as early as possible
- Continue with same drug unless ADRs
- Follow up of medication
- Drug combination for inadequate seizure control
- No sudden withdrawal
- Dose regulation by TDM
- Continue with less dose or switch to other drug during pregnancy. Do not stop!!!

Anti Parkinson's disease

- Also called paralysis agitans
- s/s include
 - Muscle rigidity
 - Tremor
 - Hypokinesia
- Types
 - Idiopathic
 - Atherosclerotic
 - Post-encephalitic
 - Drug induced (iatrogenic)

Biochemical basis

- Imbalance between dopaminergic and cholinergic system

Objectives of treatment

- Restore dopaminergic activity
- Reduce excessive cholinergic activity

Classes of drugs

- Dopaminergic drugs
 - Central Dopamine agonists: L dopa, bromocriptine
 - Dopamine reuptake inhibitors: amantadine
 - Dopamine release facilitators: amantadine
- Anticholinergics: trihexyphenidyl, benztropine

Levo dopa

- Precursor of dopamine
- Given together with peripheral dopa decarboxylase inhibitor: carbidopa
- Produces on(dyskinesia)-off (akinesia)phenomena
- Treated by drug holiday
- D/Is
 - Levodopa + MAOI= increased BP
 - Levodopa + reserpine, antipsychotics= antagonism
- C/Is: narrow angle glaucoma, psychosis, pregnancy

Bromocriptine

- Synthetic dopamine receptor agonist
- Ergot derivative
- Has longer action and less ADRs

Amantadine

- Antiviral drug
- Effective if given with levodopa
- Has central stimulant action
- Produces confusion, hallucination, loss of appetite

Dexamphetamine

- Releases dopamine/inhibits reuptake/activates dopamine receptors
- C/I: under atropine treatment

Anticholinergics

- Less effective
- Reduce only tremor and rigidity
- Used in drug induced parkinsonism
- Include
 - Trihexyphenidyl
 - Cycrimine
 - Procyclidine
 - Benztropine
 - Biperidine